

Assignment 3

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Exploratory Data Analysis

Before starting the model building procedure we implemented an EDA in order to map our data and understand fundamental information that can become useful on interpreting the outcomes of the models, the behaviour of the data, as well as if there is any necessity to conduct transformations of outlier detection procedures. Starting our first plot which is the distributions of randomly chosen features as well as the response variable Y (X.1 in the figure). We did not find the distributions of all the features since we have 88 of them (795 events and 88 variables), but already with 9 variables plus the response we are able to identify that using penalized estimators is our best option in this dataset.

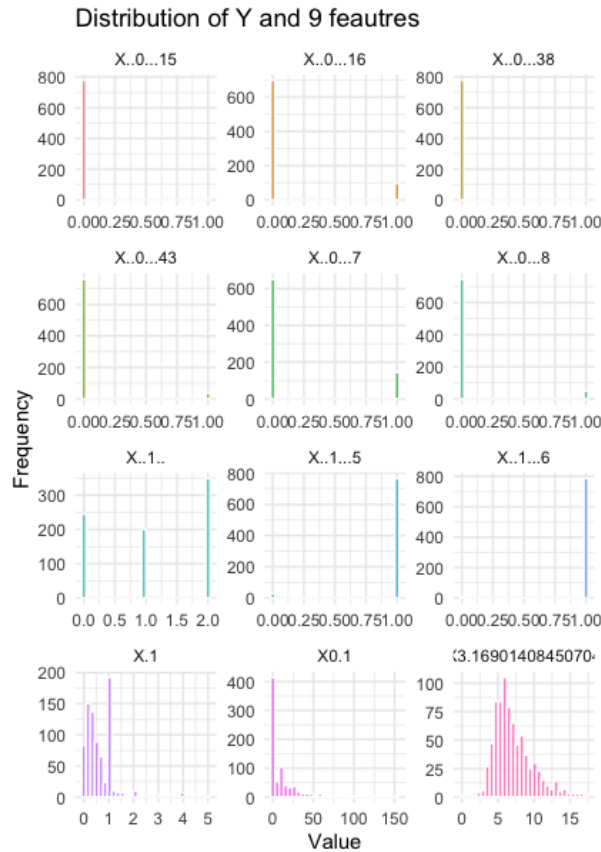


Figure 1: Histograms of 9 variables plus the response

Key Observations

From figure 1 we can report 2 basic observations about our dataset:

1. There are many sparse variables (discrete) with severe imbalances between the classes.
2. In the response (X.1) and in X0.1 we observe right-skewness, so we expect outliers and deviations from normality assumptions.

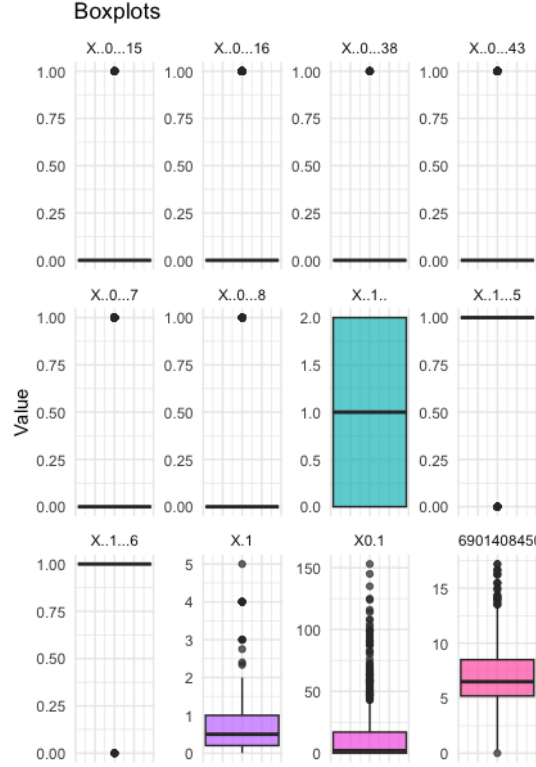


Figure 2: Boxplots of 9 variables plus the response

The boxplots further validate our initial findings, revealing significant scale disparities and an abundance of outliers. While standardisation is often recommended, it could be counterproductive in our case since many of our variables appear to play a categorical role, and applying global scaling might distort their meaning and erase the inherent structural information they carry.

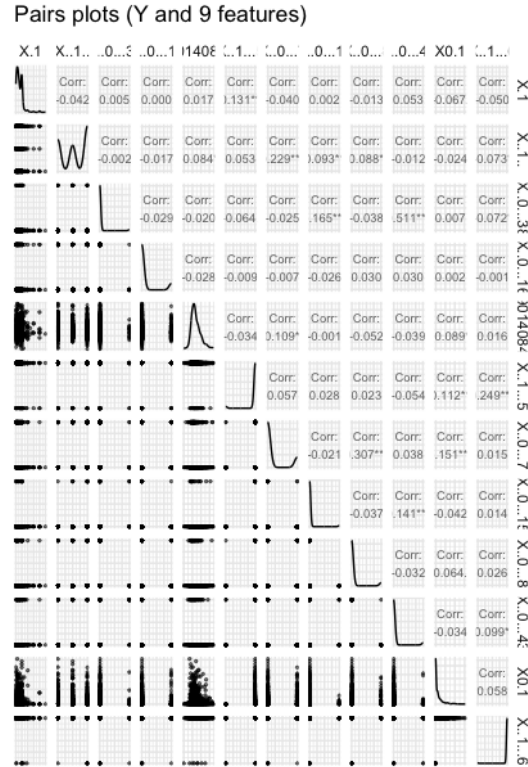


Figure 3: Pairs plot

The pairs plot analysis reveals that the majority of predictor variables exhibit negligible linear correlation, with coefficients predominantly approaching zero. Furthermore, the scatterplots demonstrate non-linear patterns, particularly in the interaction between continuous and categorical features, where standard linear assumptions are violated.

Stepwise VIF Filtering

Table 1: Variance Inflation Factor (VIF) Values for Predictor Variables

Variable	VIF	Variable	VIF
X51	2.135	X0...13	1.349
X30	1.802	X0...14	1.277
X0	1.352	X0...15	1.226
X98	5.076	X0...16	1.125
X29.102	3.838	X0...17	1.289
X26.332	9.855	X0...18	1.245
X369	6.575	X0...19	1.141
X77	44.154	X0...20	1.311
X284	26.849	X0...21	1.102
X77.1	10.870	X0...22	1.127
X15	6.460	X0...23	1.151
X76	1.584	X0...24	1.226
X79.577	4.959	X0...25	1.163
X3.169	1.194	X0...26	1.080
X20.422	4.954	X0...27	1.255
X1	1.135	X0...28	1.194
X19	14.526	X0...29	1.210
X13	3.434	X0...30	1.542
X30.1	4.468	X0...31	2.115
X20	13.567	X1...5	4.038
X94	4.796	X2..	3.730
X81	3.878	X1...6	1.653
X27	5.051	X1...7	1.766
X16	8.820	X0...32	2.496
X0.1	4.929	X1...8	4.057
X51.1	1.589	X0...33	3.856
X29.75	7.162	X1...9	2.696
X6.44	3.013	X0...34	1.191
X18	21.279	X0...35	1.270
X..0..	5.333	X0...36	1.361
X..1..	1.606	X0...37	14.375
X..0...1	3.381	X0...38	9.104
X..1...1	3.478	X0...39	20.856
X..1...2	1.304	X0...40	8.480
X..1...3	1.364	X0...41	6.216
X..1...4	1.432	X0...42	4.230
X0...2	1.149	X0...43	2.016
X0...3	2.336	X0...44	5.543
X0...4	2.664	X0...45	1.420
X0...5	4.341	X324.	16.110
X0...6	1.186		
X0...7	1.541		
X0...8	1.316		

To mitigate multicollinearity, a stepwise Variance Inflation Factor (VIF) analysis was performed to identify and remove redundant features. The resulting reduced feature set exhibits no individual variable

with extreme collinearity ($VIF \leq 10$). However, pairwise correlation analysis within this refined set still reveals localized redundancies, with correlation coefficients reaching up to 0.96. While these variables were retained to preserve the structural information of the feature space, their high pairwise correlation confirms the necessity of regularization techniques to prevent variance inflation during estimation.

Yeo-Johnson Transformation

Before reaching the last step of the preprocess stage we fitted an lm model in order to perform residual diagnostics and address if a transformation is required in order to solve any heteroscedasticity issues. Residual diagnostic plots from the initial OLS model reveal systemic violations of fundamental regression assumptions. The *Residuals vs. Fitted* and *Scale-Location* plots exhibit a clear pattern of heteroscedasticity, with the variance of residuals increasing alongside fitted values. Furthermore, the *Q-Q plot* displays significant deviations from normality at the tails, indicating non-Gaussian errors. Finally, the *Residuals vs. Leverage* plot identifies several observations with high leverage and Cook's distance, confirming that influential outliers are disproportionately affecting the model estimates (here are also 2 points that were not displayed by R because of leverage = 1). These findings provide a robust empirical justification for the subsequent data cleaning and transformation procedures.

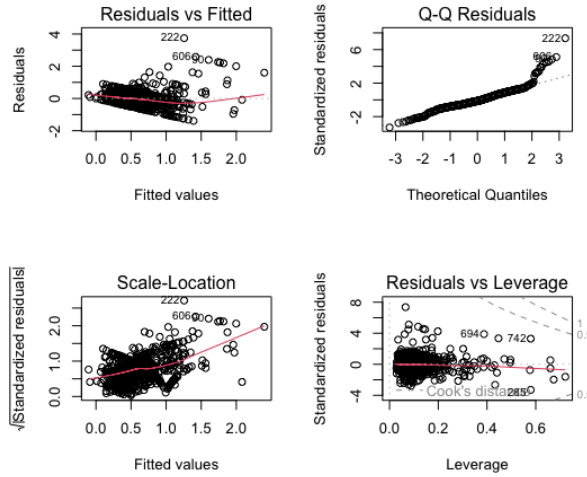


Figure 4: Residual diagnostics before Transformation

To address the non-normality and heteroscedasticity of the target variable Y , a Yeo-Johnson power transformation was implemented (Since we addressed a logarithm problem in box cox). The maximum likelihood estimation yielded an optimal parameter of $\lambda \approx -0.70$, with the chosen parameter of $\lambda = -0.5$ falling well within the Wald confidence interval $(-0.90, -0.49)$. Furthermore, the Likelihood Ratio Test (LRT) rejected the null hypothesis that $\lambda = 0$ ($p < 0.001$). However post-transformation residual diagnostics show a marked improvement in model fit. The *Q-Q plot* displays much closer to normality, and the *Residuals vs. Fitted* plot demonstrates a reduction in systemic heteroscedasticity. While residual diagnostics are not entirely ideal, working with the transformed dataset seems to be a better option here. Although we conducted a comprehensive analysis the *Residuals vs. Fitted* plot is strongly suggestive of omitted predictor bias, indicating that the simple linear functional form is insufficient to capture the underlying structure of the data. even with the usage of penalized estimators. For this reason we created a full second order formula for future experiments (We tried it in the SARIDGE model and it worked), because this corresponds to Scenario 4 from the slides (high-dimensional setting) and we are asked to check the Scenario 3.

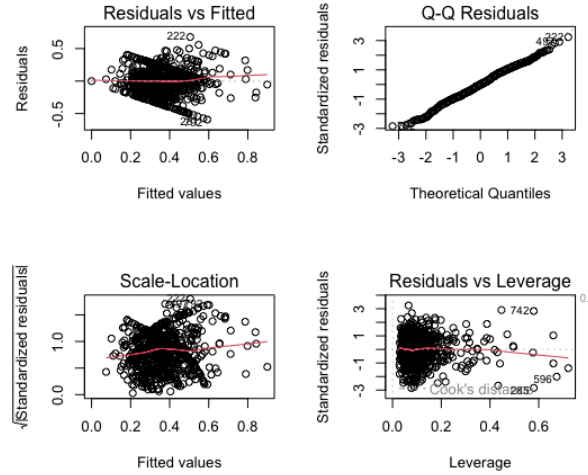


Figure 5: Residual diagnostics after Transformation

Model Building

To address the challenges of low events per variable data, collinearity, and potential outlier contamination, we implemented a set of two-stage estimators as defined in the Assignment's requirements. We also recreated the baseline penalized estimators presented in the slides such as lasso, ridge, adaptive lasso etc. Each method employs a distinct approach to feature selection and coefficient estimation.

Subsampling Adaptive Ridge (SARIDGE)

The SARIDGE estimator is implemented via a two-stage adaptive procedure:

- **Stage 1:** Ridge regression is applied to 100 subsamples (65% size) using GCV-tuning for the λ parameter. This identifies the stable feature set based on the Interquartile Range (IQR) of the coefficient estimates.
- **Stage 2:** A final Ridge regression is performed using `glmnet` with adaptive penalties. Features with unstable estimates (where the IQR of Stage 1 coefficients contains zero) are constrained to zero. The penalty factors are set inversely proportional to the median of the stable Stage 1 estimates, ensuring that the model prioritizes robust and consistent predictors.

Subsampling Least Absolute Deviations Lasso (SLAD-LASSO)

To ensure robustness against non-Gaussian errors and outliers, the SLAD-LASSO estimator utilizes quantile regression at $\tau = 0.5$:

- **Stage 1:** An Adaptive LAD Lasso is applied to 100 subsamples. The λ tuning follows the adaptive LAD Lasso framework to maintain stability in feature selection.
- **Stage 2:** Final LAD estimates are computed using only the subset of features identified as stable in the first stage, effectively filtering out noise and providing resistance to outlier contamination.

Bolasso (1 & 2)

The Bolasso framework is implemented as a two-stage bootstrap-based procedure:

- **Stage 1:** Feature selection is performed using the Bolasso framework, which identifies a reliable subset of predictors through repeated bootstrap sampling.
- **Stage 2:** Final estimation is conducted on the selected feature subset. **Bolasso1** employs the Ordinary Least Squares (OLS) estimator, while **Bolasso2** utilizes Ridge regression to obtain the final coefficients.

Evaluation Framework

The performance of these estimators was rigorously evaluated through:

1. **Monte Carlo Simulation:** Assessing the True Positive Rate (TPR), False Positive Rate (FPR), and Coefficient MSE across Gaussian and 5% contaminated scenarios.
2. **Cross-Validation:** A 5 times repeated 5-fold cross-validation measured predictive accuracy using the Mean Absolute Error (MAE) and Median Absolute Error (MedAE).

Monte Carlo Results

To rigorously assess the performance of the proposed estimators (SARIDGE, SLAD-LASSO, and Bolasso variants), we conducted a Monte Carlo simulation study. This framework allows us to evaluate model performance under controlled conditions where the true underlying parameters (β) are known.

Data Generation Process

We generated synthetic datasets with $n = 100$ observations and $p = 20$ predictors. The predictors were sampled from a multivariate normal distribution $\mathcal{N}(0, \Sigma)$, where the covariance matrix Σ was constructed using a Toeplitz structure with $\rho = 0.9$. This high correlation ($\rho = 0.9$) introduces a controlled degree of collinearity, testing the ability of the models to handle redundant features.

The response variable Y was generated as $Y = X\beta + \epsilon$, where the true coefficient vector β contained 8 active predictors and 12 noise predictors (sparse structure). We evaluated two error distributions (ϵ):

- **Gaussian Scenario:** Errors were drawn from $\mathcal{N}(0, 2.2^2)$, representing the ideal homoscedastic case.
- **Contaminated Scenario:** To test robustness, we introduced outlier contamination using a mixture distribution. Specifically, 95% of errors followed $\mathcal{N}(0, 2.2^2)$, while 5% followed $\mathcal{N}(0, 22^2)$, simulating heavy-tailed error distributions and influential outliers.

Performance Metrics

The models were compared across 150 Monte Carlo repetitions using the following metrics:

1. **Variable Selection Performance:** True Positive Rate (TPR) and False Positive Rate (FPR) to evaluate the model's ability to distinguish between relevant and irrelevant features, and Selection Accuracy for overall classification performance.
2. **Estimation Accuracy:** Mean Squared Error of the coefficients (Coef MSE) to measure the bias and variance of the estimated parameters.
3. **Predictive Performance:** Mean Absolute Error (MAE) and Mean Squared Error (MSE) on an independent test set of size $n_{test} = 1000$, assessing the generalizability of the models under both error scenarios.

Table 2: Monte Carlo Simulation Results: Gaussian Errors

Model	TPR	FPR	Sel. Acc.	Coef MSE	MAE	MSE
Bolasso1 (OLS)	0.7483	0.6683	0.4983	0.4692	1.9532	6.0057
Bolasso2 (Ridge)	0.7292	0.6817	0.4827	0.3200	1.8992	5.6719
OLS	1.0000	1.0000	0.4000	0.5522	1.9679	6.0916
Best Sub	0.2883	0.0389	0.6920	0.2625	1.8671	5.4858
Lasso	0.4033	0.2894	0.5877	0.3318	1.8932	5.6267
Ridge	1.0000	1.0000	0.4000	0.4012	1.9063	5.7043
Elastic Net	0.4825	0.3706	0.5707	0.3324	1.8847	5.5869
Ad. Lasso	0.3717	0.1972	0.6303	0.3581	1.9108	5.7385
Ad. El. Net	0.4108	0.2350	0.6233	0.3174	1.8988	5.6704
Ad. LAD Lasso	0.3650	0.1617	0.6490	0.3009	1.8944	5.6532
SARIDGE	0.6408	0.5339	0.5360	0.3510	1.9026	5.6989
SLAD-LASSO	0.5250	0.3328	0.6103	0.3950	1.9571	6.0318

Table 3: Monte Carlo Simulation Results: Contaminated Errors

Model	TPR	FPR	Sel. Acc.	Coef MSE	MAE	MSE
Bolasso1 (OLS)	0.4150	0.3522	0.5547	1.3235	3.1244	32.7329
Bolasso2 (Ridge)	0.4033	0.3189	0.5700	0.5836	2.8707	30.9104
OLS	1.0000	1.0000	0.4000	3.0843	3.4507	35.8923
Best Sub	0.1350	0.0628	0.6163	0.8916	2.9664	31.0344
Lasso	0.1483	0.0972	0.6010	0.5043	2.8150	32.1959
Ridge	1.0000	1.0000	0.4000	0.5210	2.7474	29.3003
Elastic Net	0.1617	0.1283	0.5877	0.5413	2.8083	31.0931
Ad. Lasso	0.1292	0.0900	0.5977	0.6485	2.8679	31.2824
Ad. El. Net	0.1642	0.1122	0.5983	0.6177	2.8375	30.1896
Ad. LAD Lasso	0.3725	0.1844	0.6383	0.3235	2.7017	29.8433
SARIDGE	0.6433	0.6206	0.4850	0.5216	2.8357	31.0780
SLAD-LASSO	0.4983	0.3372	0.5970	0.4449	2.7659	29.9907

The simulation study demonstrates a clear trade-off between bias and robustness. While Best Subsets performs optimally under Gaussian conditions, its performance degrades in the presence of outliers. Conversely, the Ad. LAD Lasso and SARIDGE models provide superior robustness, maintaining stable predictive performance when error distributions are contaminated. This confirms that for our empirical dataset—which showed signs of skewness and outliers in the EDA—robust estimators are not just an alternative, but a necessity for reliable inference. The performance gap between the 4 new models presented (SARIDGE, BOLASSO 1 and 2, SLAD lasso) is not huge.

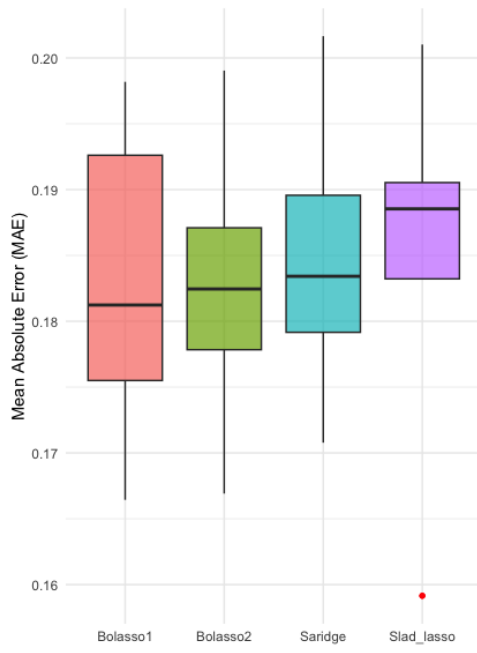
Real Data Experiment (DFR.csv)

Table 4: Comprehensive Cross-Validation Results (5-Repeated 5-Fold)

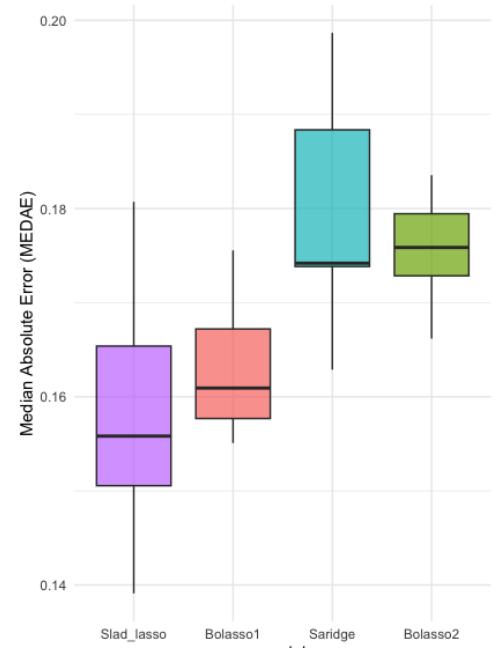
Model	Mean MAE	SD MAE	Mean MEDAE	SD MEDAE
BestSub*	0.17993	0.00807	0.16667	0.01307
Lasso	0.18026	0.00818	0.17583	0.01236
Ad.Lasso	0.18050	0.00828	0.16849	0.01275
El.Net	0.18097	0.00861	0.17709	0.01215
AENet	0.18119	0.00896	0.16942	0.01276
Ad.LAD.Lasso	0.18140	0.00804	0.17572	0.01346
Bolasso1 (OLS)	0.18316	0.01053	0.16281	0.00563
Bolasso2 (Ridge)	0.18262	0.00987	0.17539	0.00569
SARIDGE	0.18487	0.00959	0.17912	0.01181
SLAD-LASSO	0.18558	0.01312	0.15755	0.01174
Ridge	0.18371	0.00836	0.18101	0.00906
OLS	0.19234	0.00873	0.17135	0.01138

* BestSub was implemented using Forward Stepwise Selection due to computational constraints.

The cross-validation results (Table 4) demonstrate a high degree of convergence across all penalized estimators, with Mean Absolute Error (MAE) values tightly clustered between 0.179 and 0.192, significantly outperforming the standard OLS model. While Forward Stepwise Selection achieves the lowest Mean MAE, indicating strong performance in capturing the average linear structure, the robust estimators—specifically SLAD-LASSO and Bolasso—outperform it in terms of Median Absolute Error (MEDAE) and predictive stability. This suggests that while all proposed models successfully address the high-dimensional nature of the feature space, the robust frameworks provide a more resilient predictive performance against the distributional irregularities identified during EDA. Ultimately, the marginal differences in performance metrics confirm that the dataset’s abnormalities were successfully resolved, yet the gains in MEDAE and stability justify the use of these advanced estimators over traditional selection methods for ensuring reproducible and reliable inference.

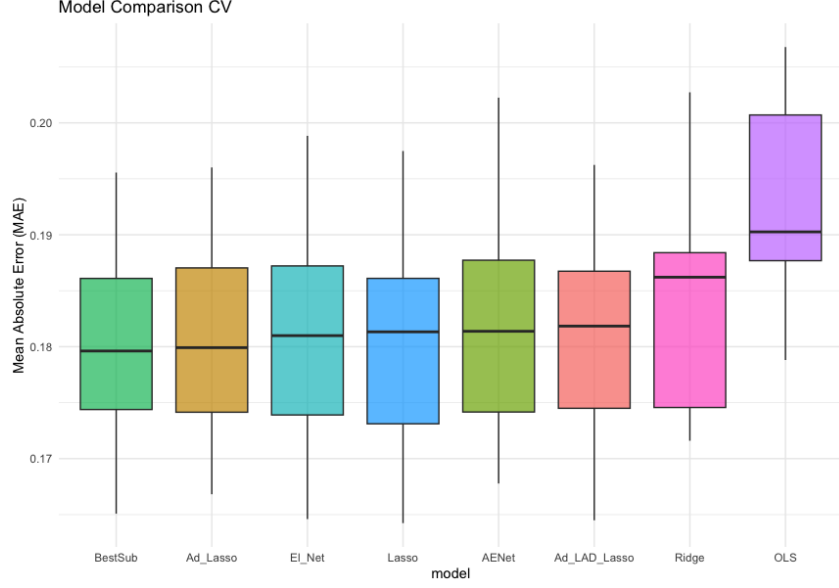


(a) Advanced models MAE

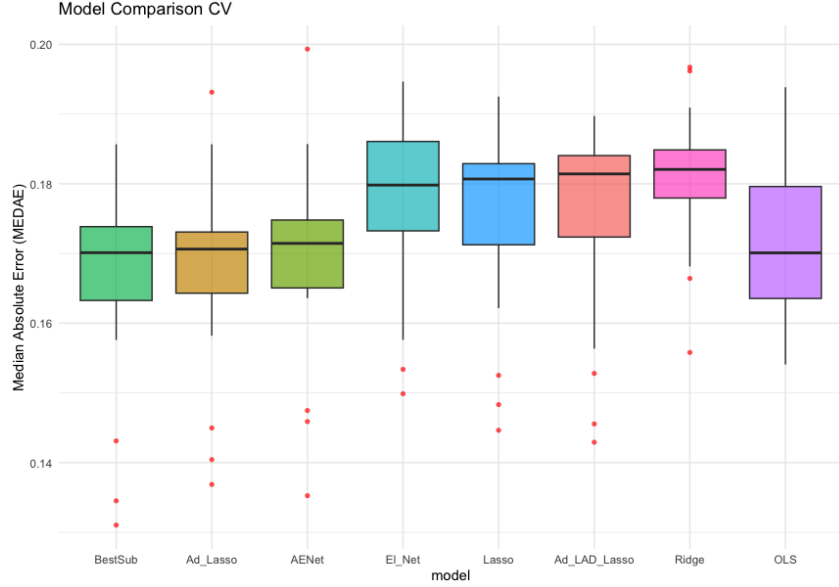


(b) Advanced models MEDAE

Figure 6: Advanced models Boxplots



(a) Baseline models MAE



(b) Baseline models MEDAE

In conclusion, while the advanced estimators demonstrated outstanding performance in terms of MEDAE and predictive stability, we must acknowledge that the performance gap between these methods and conventional ones—including OLS—was, in reality, negligible. Although the initial exploratory data analysis suggested a potentially problematic structure, the dataset proved to be well-behaved and did not represent the "worst-case scenario" we had anticipated. Regarding the omitted predictors, the consistency across our robust models confirms that their exclusion was statistically sound, further validating our feature selection process. Ultimately, while standard methods were surprisingly competitive here, our proposed frameworks provide a more resilient safety net and a more rigorous selection mechanism, which would prove essential in more volatile scenarios, such as high-dimensional settings where the number of predictors exceeds the number of observations ($p > n$).

Note: llm was used for debugging (column name management and plot improvements) and expression management